Performance and Storage Overhead Analysis of Column-Level Database Encryption

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Abstract

Data encryption is widely used around the world with the primary objective of protecting sensitive data or information stored in modern database systems. However, encryption introduces additional computational and storage overhead that have the potential to affect database performance, specifically as data volume increases. This research evaluates the performance and storage impact of column-level database encryption. This evaluation is conducted by comparing encrypted and plaintext database operations within a controlled experimental environment.

The experiments were conducted by using an Oracle APEX environment with the total dataset sizes of 10,000, 20,000 and 30,000 records. Both read-intensive and write-intensive operations, this includes select queries, bulk inserts and updates, were all executed on encrypted and unencrypted tables. Different measurements were recorded and analysed such as execution time and storage consumption to assess scalability behaviour and encryption overhead. The findings show that encryption introduces negligible overhead for read-intensive queries, while write-intensive operation experience measurable performance degradation. In addition, encrypted tables consume more storage space which becomes increasingly significant at larger dataset sizes.

Keywords— Database Encryption, Performance Overhead, Transparent Data Encryption (TDE), Column-Level Encryption, AES, Query Latency, Storage Overhead, Scalability, Oracle Database

I. INTRODUCTION

The volume and amount of sensitive information stored in database management systems has increased due to the growing dependence on digital systems and methods of storage. Confidential data such as personal records, financial information, and important business data are often contained in databases, which makes them a sough after target for unauthorised access and data breaches. Hence, database encryption has become an important mechanism for the protection of data and making sure that security and privacy regulations are complied.

However, database encryption does introduce additional data processing during data access and modification, despite its security benefits. Database performance is affected by encryption and decryption operations, specifically in systems which process large datasets and frequently perform write operations. It is essential to understand the implications on performance of encryption for administrators of databases and system designers when balancing for security needs against system efficiency.

Encryption overhead does vary depending on the characteristics of the workload, as suggested by previous studies. Minimal performance penalties occur during read-intensive operations, whereas write-intensive operations could show higher overhead due to the encryption processes. Additionally, more storage space is occupied due to overhead, which will affect the scalability of the system when in environments with resource constraints. Due to these factors, an evaluation of encryption performance under realistic scenarios and conditions is needed.

A. Problem Statement

The impact on database performance and storage efficiency is not always clearly understood in practical environments although database encryption enhances data security. Many different organisations implement encryption without ever fully evaluating the potential performance degradation and storage overhead introduced, specifically as data volume increases overtime. There is insufficient experimental analysis that quantifies the performance trade-offs of column-level encryption across different workload types and dataset sizes within managed database platforms.

B. Significance of the Problem

Excessive encryption overhead can degrade a system's responsiveness, increase the total amount of resources needed and limits scalability. Understanding the performance and storage implications of database encryption is important in order to suit deployment and informed system design. This study provides valuable insight that assists database administrators and system designers in making informed decisions regarding system design and planning.

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C. Objective

This research's main objective of this study is to evaluate the overall performance impact and storage implications of column-level database encryption through controlled experimentation. The study aims to assess the performance impact of encryption on read-intensive database operations particularly, and also to analyse the overhead that is introduced during write-intensive operations such as bulk inserts and updates, examine the scalability behaviour of encrypted databases as dataset size increases, and lastly is to evaluate the storage requirements associated with encrypted database tables.

D. Scope of the Report

The main focuses of this research paper involve the analysis of the performance and storage cost of database encryption within the column level, inside the Oracle APEX environment. The study was performed with a predetermined workload and datasets limited up to 30000 records within a controlled experiment. To note, this study does not discuss different alternative encryption techniques, neither network level security. Useful insights and conclusions on encryption overhead are meant to be obtained from the produced results, to be applied to real-world settings.

II. Literature Review

To protect important and sensitive data, database encryption is a common and essential security mechanism and tool that helps serve this purpose. Despite its utility, the computational and storage costs introduced can affect system performance, specifically in high-output environments and when latency matters. Recent research on database encryption methods and their performance overheads will be examined in this literature review, bringing focus on CPU utilisation, query latency, expansion of storage, limitations of indexing, as well as constraints in scalability. To identify encryption techniques which provide an optimal balance between security and efficiency, understanding the impacts are very important.

A. Selection Criteria

Criteria of the articles for review were based on these factors:

1. **Relevance** – The article discusses or studies upon database encryption and their related performance overhead effects.
2. **Source Quality** – Only certain types of papers were sourced for, specifically peer-reviewed journals, academic conferences, and formal academic publications such as IEEE, Springer, ResearchGate, Elsevier indexed sources were qualified.
3. **Performance Focus** – Studies which focused on analysis through theoretical or empirical methods of latency, CPU load, memory usage, performance overhead and storage consumption.
4. **Recent publishment** – Papers published in the last 5 years were sought after, the only exception was for foundational theoretical concepts that were sourced from older foundational papers.

These factors were to make sure the research possesses scholarly quality and relevance.

B. Organisation of the Literature

To show the recurring patterns in the studies, the literature was organised into 3 separate thematic categories.

- **Encryption Architecture and Topology** – Emphasises on how the deployment model e.g. software, hardware, hybrid etc. affects performance overhead
- **Encryption performance metrics** – Measures of latency, CPU usage, memory consumption, and storage overhead through empirical methods across database management systems such as Oracle, SQL Server, MySQL, PostgreSQL and MariaDB.
- **Design considerations associated with security to performance trade-offs** – A conceptual analysis of algorithm strength, key management, granularity, and indexing behaviour.

This is to make sure that there are a cohesive flow and proper comparison between studies.

C. Review of Articles

[1] Mattsson (2005)

This particular paper studies multiple encryption topologies and then evaluates their performance costs. Topologies such as network attached devices, software encryption, and hybrid models. It was revealed that network devices cause extreme latency that are up to 100 times slower. Whereas software AES offers a more balanced performance profile. The encryption topology with the highest achieved throughput is hybrid encryption with up to 2.1M row decryptions per second. Key optimisation techniques and strategies were also identified, such as avoiding encryption on indexed columns, limiting encrypted fields, and using search structures that are accelerated. Distinct trade-offs between security strength and system efficiency were demonstrated in this foundational study.

[2] Vaheedbasha & Shaik (2020)

A comparison of Oracle TDE and non-encrypted databases across versions of 12c, 18c, and 19c were done in this study. TDE was shown to significantly increase consumption of memory with up to 9 times higher, as well as moderately increasing CPU usage, and constantly causes query latency. Disk I/o was significantly worse in the earlier versions but does improve under Oracle 19c due to AES-NI and optimised methods of caching. These results bring focus towards limitations of scalability and the huge influence of database engine version on encryption overhead.

[3] Abolade et al. (2021)

The study offers valuable insights that are transferable into cryptographic overhead despite its main focus on IPSec encrypted TCP traffic rather than databases. Payload sizes are

significantly increased under AES encryption, where 1 byte is increased to 124 bytes, latency is added, and block padding with metadata expansion reduces throughput. These described behaviours reflect database encryption overhead, specifically on page-level AES encryption where expansion of ciphertext and padding results in increased storage and overall slower I/O performance.

[4] Carvalho, Sá & Bernardino (2025)
This paper undergoes a comparative evaluation of Oracle, MySQL, and SQL Server under AES-128/192/256, concluding that TDE always causes execution delays, and that SQL server achieves the best stability and scalability overall. Severe I/O penalties are caused by Oracle at larger scale factors, with up to a 4261% increase in disk time, while inconsistent CPU and latency behaviour is shown with MySQL. The fastest method is AES-128, whereas AES-256 demonstrates slowdowns with it being 427% slower than AES-128 at the 10GB scale, specifically in Oracle. As a whole, this paper shows and supports differentiation in algorithm-based performance.

[5] Shmueli et al. (2009)
Occurrences of encryption overhead at the architectural level is explained in this theoretical survey. Four specific challenges were given focus to, which were computational cost, indexing lost, storage expansion, and complexity of integration. It was concluded that page-level encryption has the lowest performance cost but weakest fine-grained security, after the authors had compared encryption granularity at the cell-level, row-level, page-level, and table-level. It was also shown that significant query latency can be resulted from encrypted indexes forcing full table scans.

[6] Madyatmadja et al. (2021)
This article shows quantitative evidence of encryption overhead focusing on SQL server TDE. It was shown that there was a 2 to 7% degradation, 8% CPU overhead, 2 to 4% increase in memory usage, and 10 to 18% in backup time penalty. Not only does encryption affect runtime query performance but also administrative tasks such as backups, as reinforced by the study. The authors note that despite TDE overhead being quite moderate, it scales and becomes more severe under high load.

[7] Gomes et al. (2021)
An evaluation of 3 encryption strategies in PostgreSQL was done in this study, which were PGP, AES, and AES with indexing. It was shown that PGP had a result of a 1.12GB insertion that lasted 8 hours, making it impractically slow. AES has an estimated 400% insertion overhead, while AES with indexing produced close to 9 times slower performance. Not only that, but query performance also declined, and database size inflated up to 4 times. This paper emphasises and shows that stronger encryption and index maintenance increases overhead by a lot, making AES the only practical approach.

[8] Istifan & Makovac (2022)
In this benchmarking study, OLTP workloads were used to evaluate AES-128/192/256 for MySQL and MariaDB. Significant performance drops were shown in MySQL with AES-256, whereas MariaDB has only slight differences between key sizes. Interestingly enough, engine-level caching effects make it so that occasionally encrypted databases outperform non-encrypted databases. AES block padding and metadata introduce storage overhead, while at higher concurrency levels, scalability tests display declining efficiency. These study findings bring focus to DBMS dependant encryption behaviour.

D. Comparison Table

Table 2.1 Comparison Table of Literature Review

Reference	Problem Statement	Methodologies	Key Findings
[1] Mattsson	High overhead from encryption topologies; need for balanced model	Compared NAED, software AES, hybrid encryption	Hybrid fastest / NAED slowest / AES efficient / indexing encryption costly
[2] Vaheedbasha & Shaik	Assess TDE overhead in Oracle	CPU/RAM/I/O benchmarks across versions	RAM ↑ 9×; CPU ↑ / latency ↑ / I/O improved in 19c
[3] Abolade et al.	Measure IPSec + AES overhead	TCP throughput and latency testing	Size ↑ drastically / latency ↑ / throughput ↓ / padding causes irregular costs
[4] Carvalho et al.	Compare AES key sizes on DBMS performance	TPC-H benchmarking on Oracle/MySQL/SQL Server	AES-256 worst / SQL Server best scaling / Oracle severe I/O overhead
[5] Shmueli et al.	Explain encryption design trade-offs	Theoretical analysis	Indexing breaks / cell-level slow / page-level fastest / security–performance trade-offs
[6] Madyatmadja et al.	Measure SQL Server TDE overhead	Load, stress, backup testing	Latency ↑ 2–7% / CPU ↑ 8% / backup ↑ 15%
[7] Gomes et al.	Evaluate encryption cost (PGP vs AES)	PostgreSQL insertion/query/storage tests	PGP too slow / AES 4–5× overhead / index + AES → 9× slower / storage ↑ 4×

[8] Istifan & Makovac	Benchmark AES key sizes in OLTP	MySQL/MariaDB TPM/NOPM tests	AES-256 worst in MySQL / MariaDB stable / occasional encrypted > unencrypted

E. Literature Review Findings, Gaps, Opportunities

Key Findings

- Overhead is always present when there is encryption, however the severity is dependent on the algorithm's strength, architecture of the DBMS, and size of data.
- Measurable CPU, latency, and I/O penalties still happen despite page-level encryption (TDE) offering reasonable and practical performance.
- Security performance trade-offs are to be considered, as stronger algorithms such as AES-256 and PGP greatly increase costs.
- Full table scans and increased latencies and forced when indexing encrypted values, thus making them a major bottleneck.
- Storage overheads can be ranged from moderate, such as TDE encryption, to extreme, such as AES with keys or PGP.
- The implementation of DBMS should be of consideration, as SQL server deals with encryption efficiently, whereas Oracle and MySQL have noticeable overhead at scale.

Gaps identified

- There is a limited amount of research done on the performance overhead of multi-tenant cloud DBMS encryption.
- Performance of degradation over a long period of time is not as widely studied and documented upon.
- Real-time query systems where latency is important, is only analysed by a few studies.
- Lack of research done on the comparison between different encryption granularities that are under similar workloads.

Opportunities for Future Research

- An implementation of an adaptive or selective encryption mechanism that only applies encryption when it is necessary.
- Reduce table scan overhead by developing index friendly encryption schemes.
- Combine page-level encryption efficiency with granular security through the exploration and development of hybrid encryption models.
- Perform benchmark tests of encryption overhead within distributed and cloud databases.

F. Summary of Literature Review

The literature reviewed can lead to the conclusion that although encryption is essential for the safeguarding and the protection of sensitive data, it will inevitably introduce performance overhead and penalties across the CPU load, memory usage, disk I/O, and storage consumption. Across all mechanisms compared, AES remains as the most practical algorithm that balances security and efficiency, despite the overhead present with key length and indexing complexity. TDE does offer a practical form of compromise but is still not free from overhead. The problem and challenges are still the same, which is to find an encryption method that provides robust security whilst minimising computational as well as storage costs.

This creates the basis for the current research, which is to evaluate different multiple encryption techniques and to study their performance impact, as well as to identify configurations which offer optimal security and efficiency balance.

III. RESEARCH METHODOLOGY

This part of the research paper describes the methodology used to evaluate the performance and storage overhead of column-level database encryption. Plaintext and encrypted database operations were compared using a controlled experimental method with different dataset sizes. It targets to measure execution time, storage impacts and scalability behaviour while also addressing the constraints caused by Oracle APEX platform. All SQL and PL/SQL scripts with their outputs used in this research are publicly available at the GitHub repository.

A. Research Design

The research uses a quantitative and comparative experimental design to evaluate the performance overhead of database encryption. Two database configurations were evaluated. A baseline configuration that uses plaintext tables and another configuration which is an encrypted type that uses column-level encryption. Both configurations were tested under identical conditions to ensure consistency and fairness. Different encryption status act as the independent variable, while execution time and storage size were treated as dependent variables. Dataset size served as a scalability factor. All experiments were conducted within the same Oracle APEX environment to eliminate variability due to hardware or system differences.

B. Steps Involved in the Methodology

1) Identification of Research Objectives:

The primary objective of this study is to measure the performance impact of column-level encryption on read-intensive database operations. Other than that, performance overhead of encryption is also evaluated on write-intensive operations such as bulk inserts and updates. Scalability trends are assessed by observing performance changes as dataset sizes increases. Lastly, storage overhead introduced by encrypted columns were analysed. These objectives were derived from gaps identified in the literature review regarding practical performance trade-offs of database encryption.

2) Literature Review and Analysis

a) Finding Information / Resources:

The sources included are relevant scholarly publications, conference papers and also technical documents related to database encryption and performance evaluation. The sources were specifically selected based on relevance to database-level encryption, empirical performance analysis and scalability considerations.

b) Techniques used for Literature Review:

A comparative synthesis approach was used to identify common findings across studies, particularly regarding read against write overhead, scalability behaviour, and storage implications. The experimental variable and performance metrics used in this study were chosen with consideration for the reviewed literature.

3) Data Collection

Synthetic datasets were generated to ensure consistency and repeatability across experiments. Three dataset sizes were used which are 10,000, 20,000 and 30,000 records. For each dataset size there are identical data distribution that are maintained for both plaintext and encrypted tables. By keeping track of each query's elapsed execution time, execution time data was automatically collected. For every test scenario, multiple attempts were run, and the average execution times of the runs were included in the study to reduce the total impact of various fluctuation.

4) Experimental Setup

All experiments included in the study were all conducted using the Oracle APEX SQL Workshop environment. Column-level encryption was implemented using Oracle's own built-in encryption mechanisms due to security restrictions that prevent the use of user-defined cryptographic packages in managed APEX environments. Execution time was also measured using Oracle's built-in timing functions, which provided results up to centisecond-level resolution. Test queries were executed sequentially but not at once to avoid interference from parallel processing behaviour. Storage usage was measured by querying database segment information to get the space that is consumed by both plaintext and encrypted tables. After processing, outcomes are imported into programs like Excel and Sheets where the numbers were turned into clear summaries, easier to visualise, and graphs were also created to show results and reveal differences through visual contrast.

5) Implementation of Solutions

A predetermined set of database operations were carried out on both plaintext and encrypted tables as part of the experimental implementations. These operations included are point select queries, bulk insert operations, bulk update operations, joining queries that involves transactional data and range-based select queries. Each of these operations were executed multiple times for both configurations, and results generated were logged into a performance results table for subsequent analysis. Baseline tests were completed before encrypted tests to maintain consistent execution order.

6) Evaluation and Comparison

For performance evaluation and comparison, it was conducted by comparing different average execution times between the two configurations which are plaintext and encrypted across all dataset sizes. About the scalability, it was assessed by looking at performance trends as dataset size increased from 10,000 to 30,000 records. Moreover, the storage consumption factor was evaluated by comparing the different disk space usage of plaintext or encrypted configuration. Lastly, graphs were also generated and used to visualise execution time trends and the total encryption overhead.

7) Validation of Results

Each of the test case were repeatedly executed and averaged to ensure the validity of the results. Consistency across multiple runs was verified before inclusion in the final dataset. Some read-intensive operations recorded to zero milliseconds due to the centisecond resolution limit of the timing mechanisms; these values show that execution times below the measurable threshold rather than the true zero-cost execution. The observed trends were also cross validated against findings from existing literatures to ensure theoretical consistency.

C. Tools and Technology Used

1) Tools

The tools that were used are Oracle APEX SQL workshop and Microsoft Excel or Google Sheets for data analysis and visualization.

2) Software

The software used are Oracle Database and the built-in Oracle column-level encryption features.

D. Flow Diagram or Architecture

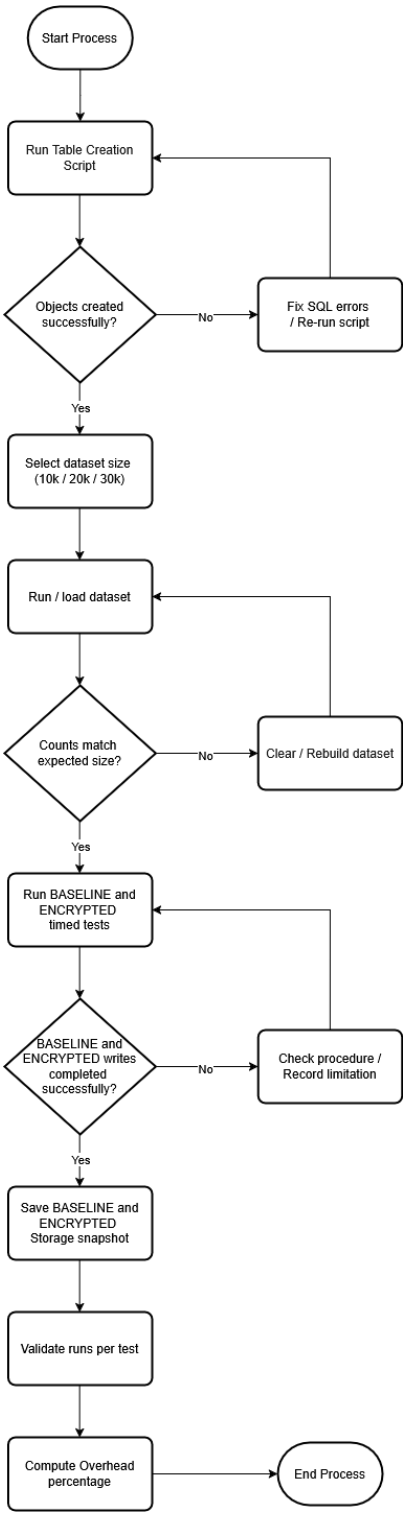


Fig. 3.1. Flowchart of Sequence of Steps Followed

E. Challenges and Limitations

A significant limitation of this study involved the Oracle APEX environment restricting access to user-defined cryptographic packages which necessitates the use of built-in encryption mechanisms. Additionally, the timing resolution limited the calculations of extremely fast read operations, this results in some near-zero execution time values. In addition, storage quota limitations within the APEX environment prevented the action of encrypted bulk insert operations at the largest dataset size from being completed. This is not considered as a methodological error, but a practical real-world limitation related to the larger storage footprint of encrypted data.

F. Summary

This methodology provides a controlled and systematic approach for evaluating the performance and storage impact of column-level database encryption. Despite there being environmental constraints, the experimental design has enabled a meaningful comparison between plaintext and encrypted configurations, this supports valid conclusions regarding encryption overhead, scalability behaviour and storage implications.

IV. Results and Discussion

The analysis of the experimental results that were obtained from the evaluation of performance and storage overhead of column-level database encryption is presented and discussed in this section. Performance tables and graphical representations that were generated using spreadsheet software (Microsoft Excel) supports the findings. This discussion will refer to these tables and figures to explain the trends and behaviours observed.

A. Results

Average of AVG_MS	Column Labels		
Row Labels	**BASELINE**	**ENC**	**Grand Total**
10000	31.34	49.3	40.34
20000	88.67	51	69.83
30000	71.67		71.67
Grand Total	**63.89**	**50.2**	**58.4**

Table 4.1. Average execution time of Baseline and ENC operations

The average execution time of baseline and encrypted database operations across dataset sizes of 10,000, 20,000, 30,000 records is summarised in Table 4.1 above. As indicated by the results, read-intensive operations such as point select and range select queries will consistently record very low times of execution across all dataset sizes. In several cases, zero millisecond execution times were recorded, meaning that its actual execution duration is below the centisecond resolution of the timing mechanism, instead of representing the true zero-cost execution. To note, average execution time for encrypted database operations at 30,000 dataset records failed to be successfully recorded due to the imposed storage quota of the Oracle APEX environment. Hence, explaining the empty data entry under ENC at 30,000 records.

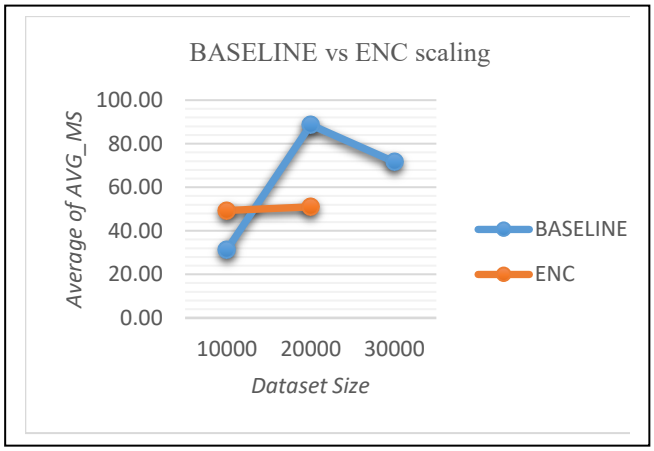


Fig. 4.1. Baseline against ENC scaling chart

The scalability of write-intensive operations is illustrated using a line chart which plots the execution time against dataset size, as shown in Figure 4.1. Execution time increases as the dataset grows for both the baseline and encrypted configurations, according to the graph. However, for bulk insert and bulk update queries, encrypted operations consistently show higher execution times compared to plain-text executions.

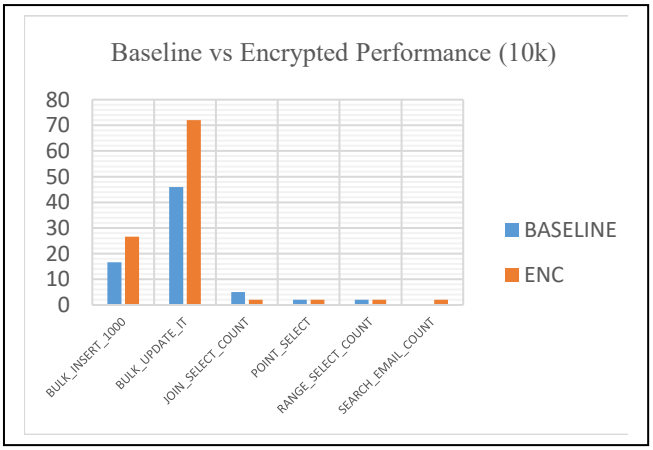


Fig. 4.2. Baseline vs Encrypted Performance (10k)

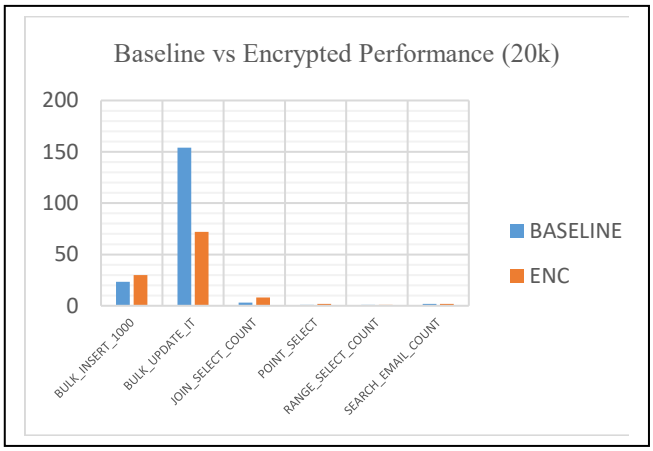


Fig. 4.3. Baseline vs Encrypted Performance (20k)

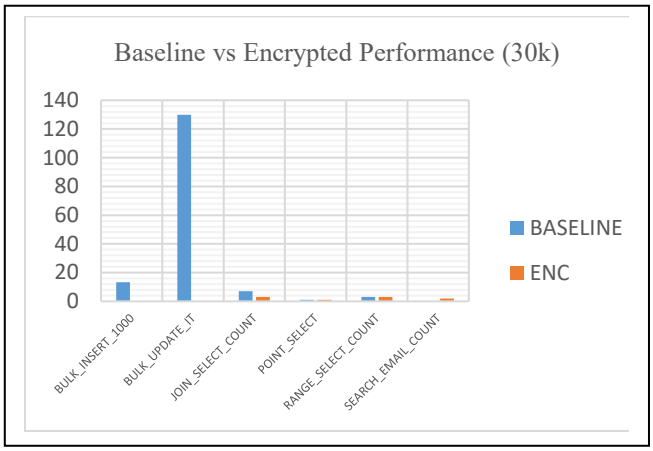


Fig. 4.4. Baseline vs Encrypted Performance (30k)

Grouped bar charts comparing baseline and encrypted execution times within datasets of 10,000, 20,000, and 30,000 is shown in Figure 4.2 to 4.4. It is clearly shown that there is a performance gap between plain-text and encrypted tables for write-intensive operations, whereas there are minimal differences for read-intensive operations.

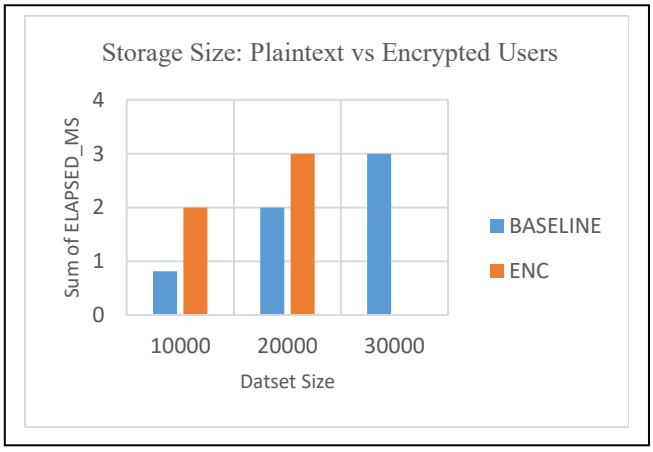


Fig. 4.5. Storage Size: Plaintext vs Encrypted Users

DATASET_SIZE	ENC_MODE	TEST_NAME	ELAPSED_MS	NOTES
10000	BASELINE	STORAGE_TRANSACTIONS_MB	2	baseline storage
10000	BASELINE	STORAGE_USERS_MB	0.81	baseline storage
10000	ENC	STORAGE_TRANSACTIONS_MB	2	encrypted storage
10000	ENC	STORAGE_USERS_MB	2	encrypted storage
20000	BASELINE	STORAGE_USERS_MB	2	baseline storage
20000	BASELINE	STORAGE_TRANSACTIONS_MB	3	baseline storage
20000	ENC	STORAGE_USERS_MB	3	encrypted storage
20000	ENC	STORAGE_TRANSACTIONS_MB	3	encrypted storage
30000	BASELINE	STORAGE_USERS_MB	3	baseline storage
30000	BASELINE	STORAGE_TRANSACTIONS_MB	5	baseline storage
30000	ENC	STORAGE_USERS_MB		encrypted storage
30000	ENC	STORAGE_TRANSACTIONS_MB		encrypted storage

Table 4.2. Plaintext vs Encrypted Users in Storage Size

Figure 4.5 and Table 4.2 present the storage usage results, which shows that encrypted tables utilise more storage space than plaintext tables, with the difference increasing as the size of the dataset grows. Encrypted bulk insert operations could

not be completed at the largest dataset size of 30,000 records, due to the Oracle Apex environment's-imposed storage quota limitations. However, this limitation does not affect the validity of the execution time results taken for the smaller dataset sizes.

B. Comparative Analysis and Discussion

By comparing Table 4.1 and Figure 4,1 to 4.4, it is revealed that encryption overhead is heavily affected by and dependant on the type of workload. As shown in Figure 4.2, read-intensive operations have execution times of encrypted and plaintext tables that are nearly the same. This signifies that decryption overhead has a slight but minimal impact on performance for selective or indexed reads during query execution.

However, in contrast to read-intensive operations, consistent performance degradation is shown with write-intensive operations under encryption. The gap between baseline and encrypted execution time is widened as the size of the dataset increases, as seen in Figure 4.1. This can be explained by the added cost of computational power needed to encrypt the data during insert and update operations, that increases in significance at the larger data volumes.

Comparison of storage size of Plaintext against Encrypted users in Figure 4.5 and Table 4.2 focuses on the impact of encryption on the resources of the system. Ciphertext expansion and encryption-related metadata causes encrypted tables to utilise and occupy more storage space. As a result, the allocated quota of Oracle APEX's environment was exceeded due to the storage overhead at the 30,000-record dataset size. This led to the failure in completing encrypted bulk insert operations. As a whole, this particular outcome demonstrated a real scalability constraint related to encrypted data storage within a controlled environment, instead of it being a limitation of the encryption technique itself.

C. Overall Discussion

To summarise what has been discussed so far, it has been shown that column-level encryption offers resilient protection of data with negligible impact on read performance, shown by the execution times that are consistently low in Table 4.1 and Figure 4.2.

The importance of considering and balancing storage capacity is vital when utilising encryption within a constrained or managed environment, as indicated by the storage trends shown in Figure 4.5.

Despite encryption overhead being manageable in moderate dataset sizes, the failure of encrypted bulk insert operations at the largest dataset size of 30,000 records shows how storage limitations can restrict the scalability of the system. Ultimately, it highlights the need for proper planning when using encryption to maximise efficiency, specifically in environments when resources are limited.

D. Critical Evaluation

Several limitations should be acknowledged even though the experimental results brought meaningful insights. For execution time measurements, it is subjected to the centisecond resolution of the timing mechanism, this limits the precision of measurements for very fast read operations. Moreover, these experiments were conducted within a managed Oracle APEX environment, which Oracle APEX imposes strong storage and configuration constraints that may not be present in fully managed on-premise systems.

Although there being limitations, the controlled experimental design ensures that the relative performance differences between baseline and encrypted configuration is still valid. Consistent dataset and repetitive measurements were utilised, to enhance the reliability of the observed trends.

E. Linking to Literature Review

The result of this study is consistent with conclusions that is written in the literature review prior on database encryption performances. Previous research suggested that read-intensive operation generally encounter minimal encryption overhead, meanwhile write-intensive operation have the tendency to require greater costs due to the encryption procedures.

Furthermore, the storage overhead shown in the encrypted tables aligns with previous research, emphasizing on the increased storage requirements needed with encrypted data. The practical storage constraints that are observed with increasing dataset sizes underscore the importance of accounting for storage overhead when integrating encryption within a real-world operational database system.

F. Results and Discussion Conclusion

In summary, the findings of this study have successfully shown that column-level database encryption needs to balance between distinct performance and storage. While read performance remains generally unaffected, write operation requires measurable overhead, and encrypted data needs additional storage space. These findings highlighted the importance of resource availability and the characteristics of workload when deploying database encryption. Overall, the findings offer useful insights for the performance implications of encryption and encourages well-informed decision-making for secure database system design.

V. CONCLUSION

This research investigates the performance and storage overhead of column-level database encryption through a controlled experimental comparison between plaintext and encrypted database operation. By evaluating read-intensive and write-intensive workloads across increasing dataset sizes, the study has already achieved its objectives of quantifying encryption overhead, analysing scalability behaviour, and assessing storage impact within a managed database environment.

Encrypted databases can maintain effective query performance for selective and indexed read. This is proved by showing that column-level encryption adds just little overhead that is negligible for read-intensive activities. Moreover, write-intensive methods show measurable performance deterioration due to the higher computational costs of encryption.

These results are also consistent with the literature review, in which it emphasizes the workload-dependent nature of encryption overhead and finding a balance between data security and system performance. The results further reinforce prior research indicating that encryption overhead is minimal for read but more important for write and storage consumption. This study provides practical evidence that fits the theoretical and empirical findings that are reported in previous studies.

The methodology employed in this research ensured valid comparisons between encrypted and plaintext configuration although there being environmental limitations such as timing constraints and storage quota restrictions. These limitations successfully reflected real-world development conditions and further enhances on the importance on planning for capacity and workload analysis.

As a conclusion, the comparative analysis included in this research provides a clear understanding of performance and storage trade-offs that are associated with column-level database encryption. Other than that, many meaningful insights are included in this study that contributes to the field of database management systems. This can be proved by showing how encryption can be effectively deployed while maintaining performance levels.

The output and SQL code used for the research are documented and located at our public GitHub repository.

REFERENCES

[1] U. T. Mattsson, “Database Encryption - How to Balance Security with Performance,” *papers.ssrn.com*, Feb. 25, 2005.https://papers.ssrn.com/sol3/papers.cfm?abstract_id=670561

[2] K. Natarajan and S. Vaheedbasha, “Transparent Data Encryption: Comparative Analysis and Performance Evaluation of Oracle Databases | IEEE Conference Publication | IEEE Xplore,” *ieeexplore.ieee.org*, Dec. 21,2020.https://ieeexplore.ieee.org/abstract/document/9296168

[3] O. Abolade, A. Okandeji, A. Oke, M. Osifeko, and A. Oyedeji, “Overhead effects of data encryption on TCP throughput across IPSEC secured network,” *Scientific African*, vol. 13, p. e00855, Sep. 2021, doi: https://doi.org/10.1016/j.sciaf.2021.e00855.

[4] M. Carvalho, F. Sá, and J. Bernardino, “Evaluation of the Impact of AES Encryption on Query Read Performance Across Oracle, MySQL, and SQL Server Databases,” *Cryptography*, vol. 9, no. 4, p. 77, Dec. 2025, doi: https://doi.org/10.3390/cryptography9040077.

[5] E. Shmueli, R. Vaisenberg, Y. Elovici, and C. Glezer, “Database encryption,” *ACM SIGMOD Record*, vol. 38, no. 3, pp. 29–34, Dec. 2010, doi: https://doi.org/10.1145/1815933.1815940.

[6] E. D. Madyatmadja, A. N. Hakim, and D. J. M. Sembiring, “Performance testing on Transparent Data Encryption for SQL Server’s reliability and efficiency,” *Journal of Big Data*, vol. 8, no. 1, Oct. 2021, doi: https://doi.org/10.1186/s40537-021-00520-z.

[7] A. Gomes, C. Santos, C. Wanzeller, and P. Martins, “Database Encryption for Balance Between Performance and Security,” *Journal of Information Assurance & Cybersecurity*, pp. 1–9, Apr. 2021, doi: https://doi.org/10.5171/2021.614511.

[8] S. Istifan and M. Makovac, “Performance benchmarking of data-at-rest encryption in relational databases,” *DIVA*, 2022.https://www.divaportal.org/smash/record.jsf?pid=diva2%3A1672343&dswid=2731 (accessed Dec. 04, 2025).